Krishnam Jhalani

+1 (437) 299-9437 | k2jhalani@uwaterloo.ca | LinkedIn | GitHub | Portfolio

Education

University of Waterloo — Waterloo, ON
Bachelor of Computer Science (Honours) — OOP, Algorithms (Adv.), Functional Programs (Adv.), Linux Tools — *Apr. 2029*

Work Experience

Founder & Software Engineer — Apr. 2026 – Present
What2Do — *Baltimore, MD, USA*

- Shipped **full-stack platform** in **TypeScript**, **Next.js**, **React**, **PostgreSQL** to **1,000+ users** in **48 hours**
- Architected **PostgreSQL** backend: **20+ tables**, row-level security, **WebSocket** subscriptions, **99.5% uptime**
- Built **ML recommendation engine** with collaborative filtering and **sentence-transformer** embeddings
- Designed native **iOS app** in **Swift/SwiftUI** with biometric auth, WebSocket sync, and offline caching

AI Developer — Sept. 2026 – Dec. 2026
UWaterloo Future Cities Institute — *Waterloo, ON*

- Building an **AI-powered planning tool** generating aligned **economic development plans** from **StatsCan** data
- Developing an **NLP grant writing tool** enabling Northern communities to independently draft funding applications
- Integrating **survey**, **municipal**, and **StatsCan** datasets into unified pipelines for two production tools

Software Engineer — May 2025 – Aug. 2025
Fluxion AI — *Remote*

- Shipped production features across **React/TypeScript** and **Python/Node.js** to **15,000+** daily users at **99.5%**
- Automated **ETL pipelines** across **500+ datasets** with **pandas/NumPy**, cutting overhead by **85%**
- Built **C++** and **C#** production modules; deployed microservices to **AWS** with **Docker** and **CI/CD**
- Built **Golang** network automation using **NETCONF/YANG** and **Kafka** telemetry, cutting deploy time **75%**

Business Systems Analyst — Jan. 2026 – Apr. 2026
Blake, Cassels & Graydon LLP — *Toronto, ON*

- Deployed **35+ AI agents** improving efficiency **40%** across **1,500+ users** at a national law firm in Toronto
- Led **UAT** with **50+ stakeholders** across **7 offices**; documented **100+ test cases** for full production rollout

Selected Projects

OmenAI | *Python, FastAPI, LangChain, PostgreSQL, Redis, Next.js, Docker* — GenAI Genesis Hackathon Winner

- **Fraud analytics platform**; won **Canada's largest AI hackathon** by shipping it in under **24 hours**
- Ran **Isolation Forest** across **600K+ records/query**; cut latency **15s→sub-second** via **Redis** caching
- Deployed **LLM-powered** risk explanation pipeline with **parallel async workers** and **>60%** cache hit rate

TripMind | *Python, FastAPI, PostgreSQL, pgvector, scikit-learn, Celery, Redis, React*

- **AI travel recommendation platform**: **TruncatedSVD** + **transformer embeddings** blended with dynamic alpha
- **pgvector IVFFLAT** cosine search over **10K listings**; **p95 under 300ms** at 50 concurrent users via Locust
- **Celery** async pipeline recomputes preference vectors every **10 interactions**; **A/B** framework with chi-square

LawDer | *TypeScript, Next.js, Python, FastAPI, LangChain, LangGraph, pgvector*

- **AI legal document analysis** SaaS; **Next.js** frontend, **FastAPI** backend, **10,000+ docs** at **88% accuracy**
- Built **LangGraph multi-agent** system with dynamic routing, tool use, and **self-healing retry** on failures
- Optimized **PostgreSQL** queries and **Redis** caching; cut document retrieval from **minutes to sub-second**

Engineering Teams

WAT.ai – Weatherloo Subteam — Waterloo, ON
Member, 8–9 Person Cross-Functional ML Research Team

- Built **48 regression models** on **ERA5/GFS** data; reduced **RMSE 5%** vs GFS baseline across **35,000+** pairs
- Built **React/Leaflet** visualization rendering **360×181** geospatial forecast grid with real-time playback

WatCloud — Waterloo, ON
Contributor, Student-Run Compute Cluster

- Configured **Ubuntu** servers and **multi-tenant access controls** for campus-wide **GPU/CPU** compute cluster
- Contributed to **Slurm-style** resource scheduling, **TCP/IP/DNS** config, and user permission management

Technical Skills

Languages: Python, TypeScript, JavaScript, C++, C#, Go, Swift, SQL, Java, Bash, HTML/CSS
Frameworks: React, Next.js, Node.js, FastAPI, Flask, SwiftUI, Tailwind CSS, Express.js
Tools: Git, Docker, Kubernetes, AWS, PostgreSQL, Redis, MongoDB, CI/CD, GitHub Actions, Linux
AI/ML: LangChain, LangGraph, PyTorch, scikit-learn, pgvector, pandas, NumPy, RAG, LLM APIs